Vector addition

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Add displacement vectors and use a model to explain where an object ends up.

SUCCESS CRITERIA - I can...

1. Connect tip-to-tail diagrams to signed component sums.
2. Give magnitude, units and an angle with a named reference direction.
3. Check the result against the diagram, limits and physical situation.

Core: teacher selects from 1-10. **Stretch:** 11-12 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Retrieve and distinguish

A cart moves from 0 m to +6 m in 3 s, then to +2 m in 2 s. Find average speed and average velocity for the full trip. Sketch the route and explain why the answers differ.

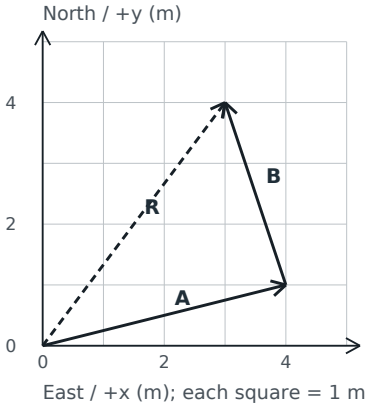
Marzano check - circle: ① ② ③

02 Explain a worked model

Reference: $R = \sqrt{(R_x^2 + R_y^2)}$.

Worked reasoning: East=+x; north=+y. R_x=4-1=3 m; R_y=1+3=4 m; R=√(3²+4²)=5 m.
θ=tan⁻¹(4/3)=53.1° north of east.

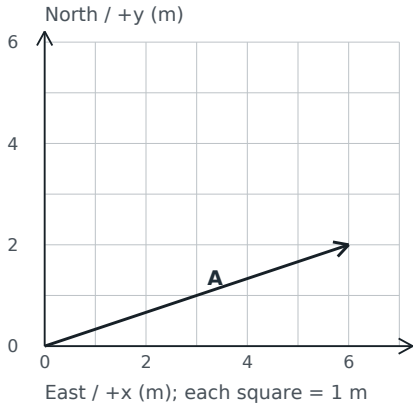
Your turn: Find B's westward component. Explain why R begins at the original start.



Marzano check - circle: ① ② ③

03 Complete, then check

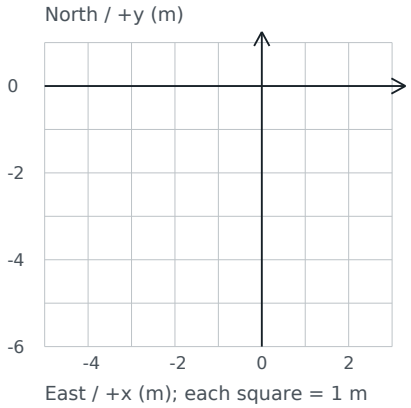
A robot makes displacement A = (6, 2) m, then B = (-4, 3) m. East is +x; north is +y. A is drawn. Add B tip-to-tail and draw the resultant from the original start. Find its components, magnitude and direction.



Marzano check - circle: ① ② ③

04 A different quadrant

A rover makes two moves: A = (2, -5) m and B = (-6, 2) m. East is +x; north is +y. Predict the endpoint quadrant, then draw the route. Calculate the resultant components, magnitude and direction; check them against the sketch.



Marzano check - circle: ① ② ③

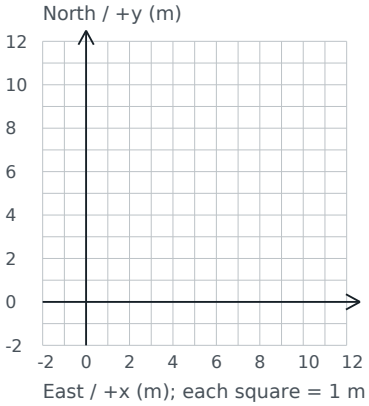
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Continue down each column.

05 Choose the correct component signs

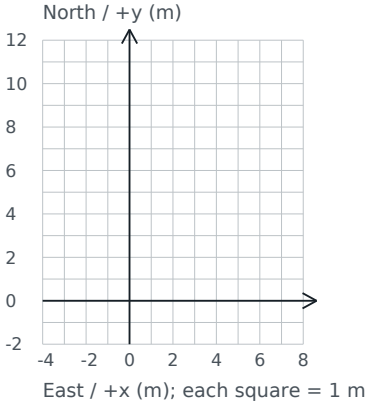
A is 10 m at 30° north of east: $A_x = 10 \cos 30° = 8.66$ m and $A_y = 10 \sin 30° = 5.00$ m.
B is 6 m at 60° north of west. Sketch both arrows, resolve B into signed components, and find the resultant magnitude and direction.



Marzano check - circle: ① ② ③

06 The angle is measured from north

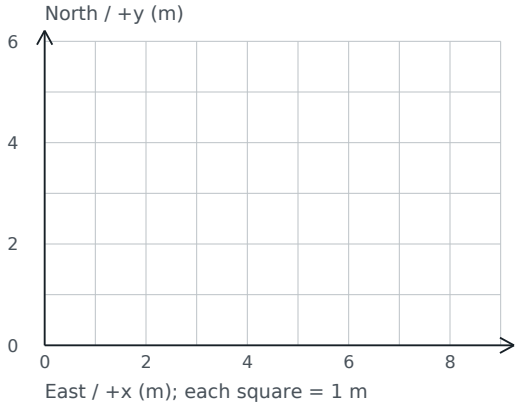
Walk 12 m at 25° east of north, then 7 m west. Sketch first and predict the resultant quadrant. Calculate components, magnitude and direction. Explain which component uses cosine and why.



Marzano check - circle: ① ② ③

07 Plan the direct return

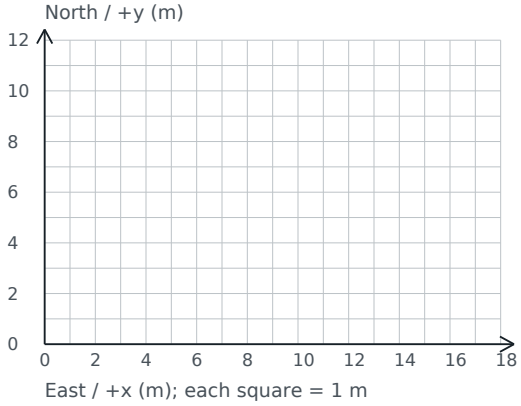
A survey team walks 8 m east, 5 m north, then 2 m west. A clear field allows a straight return to the start. Find the route distance before returning and the direct return displacement, including magnitude and direction. Draw both the outward route and the return.



Marzano check - circle: ① ② ③

08 Diagnose the reasoning

A student adds 10 m east and 10 m at 60° north of east using $\sqrt{(10^2 + 10^2)} = 14.1$ m. Identify the violated condition. Draw a correct model and use it to repair the solution, including direction.



Marzano check - circle: ① ② ③

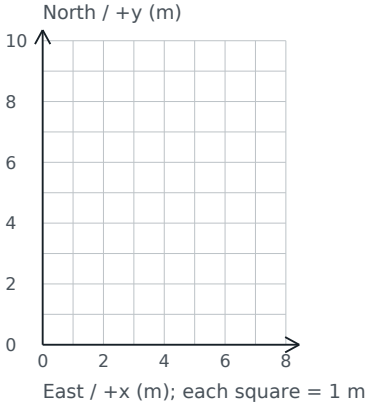
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Continue down each column.

09 Represent one trip in several ways

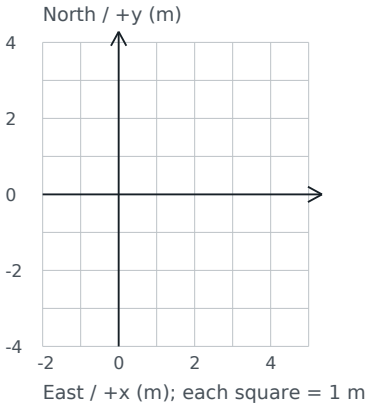
A robot moves steadily from (0, 0) m to (6, 0) m in 3 s, then steadily to (6, 8) m in another 4 s.
(a) Draw the path and resultant; find distance and displacement.
(b) Find average speed and the average-velocity vector for the full trip.
(c) Sketch x-versus-time and y-versus-time graphs. Explain how each graph records the same motion as your path diagram.



Marzano check - circle: ① ② ③

10 Exit check - solve without the worked model

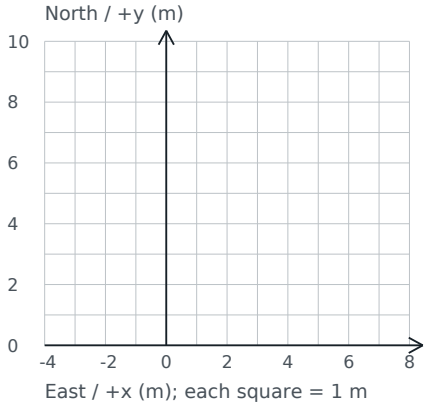
A = (4, -3) m and B = (-1, 5) m. Find R, its magnitude and direction. Choose C so A + B + C = 0. Draw a closed tip-to-tail route and give one physical or mathematical check of your answer.



Marzano check - circle: ① ② ③

11 STRETCH: Missing direction

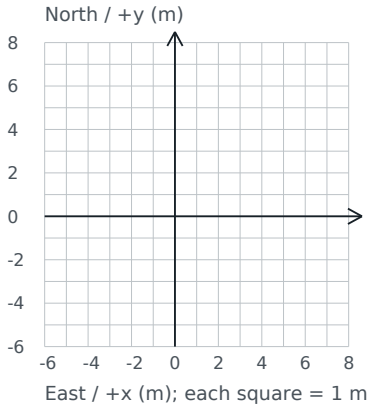
The first displacement is 6 m east. The second has magnitude 10 m. Finish directly north of the start. Find the second vector's components and direction, and the resultant. Draw the model. Explain whether more than one solution meets the north requirement.



Marzano check - circle: ① ② ③

12 STRETCH: Decide what is possible

Two displacements have magnitudes 5 m and 3 m. Test whether resultants of 1 m, 6 m and 9 m are possible using limits. Draw the 6 m case to scale. Estimate the angle between the original arrows with tails together. Explain why the tip-to-tail triangle uses a different interior angle. Optional: derive an angle-resultant relationship to check your drawing.



Marzano check - circle: ① ② ③